Foundations of Information retrieval

[Information retrieval]

1.0 Content Block

In order to effectively retrieve relevant documents by IR strategies, the documents are typically transformed into a suitable representation. Each retrieval strategy incorporates a specific model for its document representation purposes. The picture on the right illustrates the relationship of some common models. In the picture, the models are categorized according to two dimensions: the mathematical basis and the properties of the model.

2.0 Content Block

Timeline Before the 1900s 1801: Joseph Marie Jacquard invents the Jacquard loom, the first machine to use punched cards to control a sequence of operations. 1880s: Herman Hollerith invents an electro-mechanical data tabulator using punch cards as a machine readable medium. 1890 Hollerith cards, keypunches and tabulators used to process the 1890 US census data. 1920s–1930s Emanuel Goldberg submits patents for his "Statistical Machine", a document search engine that used photoelectric cells and pattern recognition to search the metadata on rolls of microfilmed documents. 1940s–1950s late 1940s: The US military confronted problems of indexing and retrieval of wartime scientific research documents captured from Germans. 1945: Vannevar Bush's As We May Think appeared in Atlantic Monthly. 1947: Hans Peter Luhn (research engineer at IBM since 1941) began work on a mechanized punch card-based system for searching chemical compounds. 1950s: Growing concern in the US for a "science gap" with the USSR motivated, encouraged funding and provided a backdrop for mechanized literature searching systems (Allen Kent et al.) and the invention of the citation index by Eugene Garfield. 1950: The term "information retrieval" was coined by Calvin Mooers. 1951: Philip Bagley conducted the earliest experiment in computerized document retrieval in a master thesis at MIT. 1955: Allen Kent joined Case Western Reserve University, and eventually became associate director of the Center for Documentation and Communications Research. That same year, Kent and colleagues published a paper in American Documentation describing the precision and recall measures as well as detailing a proposed "framework" for evaluating an IR system which included statistical sampling methods for determining the number of relevant documents not retrieved. 1958: International Conference on Scientific Information Washington DC included consideration of IR systems as a solution to problems identified. See: Proceedings of the International Conference on Scientific Information, 1958 (National Academy of Sciences, Washington, DC, 1959) 1959: Hans Peter Luhn published "Auto-encoding of documents for information retrieval". 1960s: early 1960s: Gerard Salton began work on IR at Harvard, later moved to Cornell. 1960: Melvin Earl Maron and John Lary Kuhns published "On relevance, probabilistic indexing, and information retrieval" in the Journal of the ACM 7(3):216–244, July 1960. 1962: * Cyril W. Cleverdon published early findings of the Cranfield studies, developing a model for IR system evaluation. See: Cyril W. Cleverdon, "Report on the Testing and Analysis of an Investigation into the Comparative Efficiency of Indexing Systems". Cranfield Collection of Aeronautics, Cranfield, England, 1962. * Kent published Information Analysis and Retrieval. 1963: * Weinberg report "Science, Government and Information" gave a full articulation of the idea of a "crisis of scientific information". The report was named after Dr. Alvin Weinberg. * Joseph Becker and Robert M. Hayes published text on information retrieval. Becker, Joseph; Hayes, Robert Mayo. Information storage and retrieval: tools, elements, theories. New York, Wiley (1963). 1964: * Karen Spärck Jones finished her thesis at Cambridge, Synonymy and Semantic Classification, and continued work on computational linguistics as it applies to IR. * The National Bureau of Standards sponsored a symposium titled "Statistical Association Methods for Mechanized Documentation". Several highly significant papers, including G. Salton's first published reference (we believe) to the SMART system. mid-1960s: * National Library of Medicine (NLM) developed MEDLARS Medical Literature Analysis and Retrieval System, the first major machine-readable database and batch-retrieval system. * Project Intrex at MIT. 1965: J. C. R. Licklider published Libraries of the Future. 1966: Don Swanson was involved in studies at University of Chicago on Requirements for Future Catalogs. late 1960s: F. Wilfrid Lancaster completed evaluation studies of the MEDLARS system and published the first edition of his text on information retrieval. 1968: * Gerard Salton published Automatic Information Organization and Retrieval. * John W. Sammon, Jr.'s RADC Tech report "Some Mathematics of Information Storage and Retrieval..." outlined the vector model. 1969: Sammon's "A nonlinear mapping for data structure analysis Archived 2017-08-08 at the Wayback Machine" (IEEE Transactions on Computers) was the first proposal for visualization interface to an IR system. 1970s early 1970s: * First online systems—NLM's AIM-TWX, MEDLINE; Lockheed's Dialog; SDC's ORBIT. * Theodor Nelson promoting concept of hypertext, published Computer Lib/Dream Machines. 1971: Nicholas Jardine and Cornelis J. van Rijsbergen published "The use of hierarchic clustering in information retrieval", which articulated the "cluster hypothesis". 1975: Three highly influential publications by Salton fully articulated his vector processing framework and term discrimination model: * A Theory of Indexing (Society for Industrial and Applied Mathematics) * A Theory of Term Importance in Automatic Text Analysis (JASIS v. 26) * A Vector Space Model for Automatic Indexing (CACM 18:11) 1978: The First ACM SIGIR conference. 1979: C. J. van Rijsbergen published Information Retrieval (Butterworths). Heavy emphasis on probabilistic models. 1979: Tamas Doszkocs implemented the CITE natural language user interface for MEDLINE at the National Library of Medicine. The CITE system supported free form query input, ranked output and relevance feedback. 1980s 1980: First international ACM SIGIR conference, joint with British Computer Society IR group in Cambridge. 1982: Nicholas J. Belkin, Robert N. Oddy, and Helen M. Brooks proposed the ASK (Anomalous State of Knowledge) viewpoint for information retrieval. This was an important concept, though their automated analysis tool proved ultimately disappointing. 1983: Salton (and Michael J. McGill) published Introduction to Modern Information Retrieval (McGraw-Hill), with heavy emphasis on vector models. 1985: David Blair and Bill Maron publish: An Evaluation of Retrieval Effectiveness for a Full-Text Document-Retrieval System. 1986: Donald A.B. Lindberg M.D., NLM Director, implemented a direct health professional interface to MEDLINE and other MEDLARS databases. Grateful Med, a pun on the Grateful Dead, was adapted from Microsearch, an ELHILL user interface that assembled query language prior to connecting to the NLM mainframe. After retrieving the query results, Grateful Med disconnected from the mainframe to keep search costs low. mid-1980s: Efforts to develop end-user versions of commercial IR systems. 1985–1993: Key papers on and experimental systems for visualization interfaces. Work by Donald B. Crouch, Robert R. Korfhage, Matthew Chalmers, Anselm Spoerri and others. 1989: First World Wide Web proposals by Tim Berners-Lee at CERN. 1990s 1992: First TREC conference. 1997: Publication of Korfhage's Information Storage and Retrieval with emphasis on visualization and multi-reference point systems. 1998: Google is founded by Larry Page and Sergey Brin. It introduces the PageRank algorithm, which evaluates the importance of web pages based on hyperlink structure. 1999: Publication of Ricardo Baeza-Yates and Berthier Ribeiro-Neto's Modern Information Retrieval by Addison Wesley, the first book that attempts to cover all IR. 2000s 2001: Wikipedia launches as a free, collaborative online encyclopedia. It quickly becomes a major resource for information retrieval, particularly for natural language processing and semantic search benchmarks. 2009: Microsoft launches Bing, introducing features such as related searches, semantic suggestions, and later incorporating deep learning techniques into its ranking algorithms. 2010s 2013: Google's Hummingbird algorithm goes live, marking a shift from keyword matching toward understanding query intent and semantic context in search queries. 2018: Google AI researchers release BERT (Bidirectional Encoder Representations from Transformers), enabling deep bidirectional understanding of language and improving document ranking and query understanding in IR. 2019: Microsoft introduces MS MARCO (Microsoft MAchine Reading COmprehension), a large-scale dataset designed for training and evaluating machine reading and passage ranking models. 2020s 2020: The ColBERT (Contextualized Late Interaction over BERT) model, designed for efficient passage retrieval using contextualized embeddings, was introduced at SIGIR 2020. 2021: SPLADE is introduced at SIGIR 2021. It's a sparse neural retrieval model that balances lexical and semantic features using masked language modeling and sparsity regularization. 2022: The BEIR benchmark is released to evaluate zero-shot IR across 18 datasets covering diverse tasks. It standardizes comparisons between dense, sparse, and hybrid IR models.